A particle model of hexane removal and water uptake in soybean meal desolventizing

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Abstract

In a desolventizer-toaster (DT), steam desolventization of oilseed meal couples hexane removal, water uptake and heat transfer. We formulate a conservative particle model containing free liquid hexane, binary pore gas, retained water and hexane, and a separate external water film. Component and energy jump conditions determine how the liquid-hexane interface recedes; after the core disappears, the same inventories continue to evolve in a stationary radial particle. Published thermodynamic and retention properties, together with declared transport coefficients, parameterize the model. The formulation is exercised in two prescribed baths, in which the surrounding gas history is specified rather than solved. In the first, a water-bearing soybean particle in Faner's pure-hexane vapour at 120 °C falls from 0.20 to 0.10 kg hexane/kg dry meal in 17.7 s, compared with 25.4 s measured. The source reports preheating, but not the exact initial temperature, moisture or mass-weighted particle sizes. In the second, a particle followed through a 1766.5-s prescribed DT history leaves with 17.5% wet-basis moisture, 138 ppm wet-basis hexane and a mean temperature of 105.0 °C. In both calculations, independently accumulated water, hexane and energy balances close within 4×10^{-10} in normalized units through all three regimes; whole-history mesh and time-step convergence is not yet established. The particle response is benchmarked against available literature experiments and published industrial outlet ranges, without fitting the falling-rate trajectories or DT outlet values. This common formulation provides the particle-level basis for a DT digital twin that can shorten process design and support model-based control.

Keywords: desolventization; soybean meal; n-hexane; moving-front model; sorption; food process modeling

Practical Applications. A prescribed-bath experiment lets a process engineer examine solvent loss, moisture uptake and temperature without first solving a complete desolventizer-toaster (DT). The same particle equations describe a pure-hexane drying experiment and a steam-rich DT scenario. In the pure-hexane case, the calculated particle dries faster than the measured charge; better size, exposure and temperature information is needed to explain the difference. In the DT case, the particle reaches 17.5% wet-basis moisture with liquid-water imbibition included; the uptake rate rests on one same-material measurement. The model returns signed water, hexane and energy fluxes that can later be coupled to conservative tray balances.

1 Introduction

Solvent extraction leaves oilseed meal wet with hexane. In a desolventizer-toaster (DT), live steam strips the solvent while heated trays supply additional energy (Karnofsky, 1985; Sipos & Witte, 1961; Witte, 1995). Residual solvent matters for safety and emissions; condensed water must later be removed, and the meal's thermal history affects protein quality (Kemper, 2013; Mustakas, Moulton, Baker, & Kwolek, 1981; U.S. EPA, 1995). The engineering question is how to supply enough heat and residence time for solvent removal and toasting without excessive steam use, later drying duty or damage to meal quality (Kemper, 2013; Witte, 1995). A particle model must therefore connect solvent removal, moisture uptake and heat transfer throughout the transition from externally wetted meal to a particle containing only retained material and pore gas.

These processes are coupled. Evaporation consumes energy at the particle surface or an internal liquid boundary. Incoming steam can condense, and both retained water and residual hexane contribute to storage and transport. Liquid-water imbibition also differs from vapour sorption. Experiments on soy protein isolates show that imbibition combines sorption, capillarity and interstitial retention, whereas vapour sorption describes a different hydration experiment (Jovanovich, Puppo, Giner, & Añón, 2003). Those measurements motivate treating liquid contact separately from vapour exposure; their capacities and rates cannot be transferred directly to hot, hexane-bearing extracted meal.

Published desolventization models emphasize different parts of this sequence. The superheated-hexane experiments and receding-front model of Faner, Perez, and Crapiste (2019) describe a heat-fed liquid-hexane core. The transport account of Cardarelli, Crapiste, and Mattea (2002) resolves a diffusion-controlled drying tail. Coletto, Bandoni, and Blanco (2022) couples a transient radial particle to a steady axial bed in the diffusion-controlled zone, treating the preceding flashing zone separately. Here we combine component and energy conservation at a receding liquid-hexane interface with binary pore gas, matrix retention and a separately conserved external water film. After the core disappears, the same stored components and energy continue to evolve in a stationary radial particle.

In the first of two prescribed baths, a water-bearing particle in pure superheated hexane is compared with the soybean experiment of Faner et al. (2019); pure supplied vapour does not imply a water-free meal. In the second, a particle is followed from feed through a prescribed steam-rich DT history. Prescribing the bath isolates the particle response, so the surrounding vapour composition and temperature need not be predicted. In a future coupled tray calculation, the signed water, hexane and energy fluxes would enter the gas and tray balances, which would determine the next local bath state.

The two comparisons rest on different kinds of evidence. In the Faner experiment, the measured constant-rate duty informs external heat supply; later loading and temperature readings provide the comparisons. Faner reports preheating in the apparatus and particle-size summaries, but not the exact initial temperature, initial meal moisture or a mass-weighted size histogram. The DT history is an engineering scenario; published outlet ranges provide process context rather than measurements of the same particle and operating point. Liquid-water imbibition is part of the particle formulation, and its transport coefficients are declared rather than measured. Numerical verification and complete time integration make these assumptions inspectable; independent measurements are needed to establish their predictive accuracy.

The paper has two goals: to state one conservative particle formulation and to exercise it in both baths. Section 2 defines the particle equations and phase regimes, and Section 3 their numerical integration, admissibility checks and verification. Section 4 presents the two experiments, Section 5 discusses their meaning for DT practice and the measurements needed next, and Section 6 concludes.

The Supplementary Material contains detailed source readings, parameters, numerical evidence and the separate scalar Fickian comparison with the earlier Faner measurements.

2 Particle model

This section states the particle model that the rest of the paper exercises: the phase regimes of one particle, the balances in each region, the conditions that close the moving front and the properties that enter them. Readers mainly interested in the process results can read Section 2.1 and then continue with Section 4. Table 1 gives a compact map of the governing balances and closures. The supporting equation tables in Sec. S14 distinguish literature correlations, relations derived here and diagnostic comparisons; numerical inputs remain in Table 2.

2.1 Particle geometry, inventories and phase regimes

The model follows one particle through three regimes, defined by where liquid hexane remains. Hexane is stored as external free hexane, hexane in a liquid-hexane core, retained hexane and pore gas. External free hexane wets the surface in regime A. Its depletion allows the liquid-hexane core to recede in regime B, leaving a dry shell. Throughout this paper, *dry shell* means depleted of free liquid hexane, not free of water or retained hexane. Regime C begins when the liquid-hexane core disappears. Steam can form an external water film, from which *liquid-water imbibition* transfers water into the particle's retained-water inventory. Retained water includes effective matrix and interstitial retention; it does not imply that all water is molecularly bound.

We approximate the particle by a porous sphere of fixed radius R_p. The liquid-hexane core has radius $s(t)$, with $0 \le s \le R_\mathrm{p}$; the dry shell occupies $s < r < R_\mathrm{p}$. This geometry resolves radial heat and mass transport with one length scale, as in the particle descriptions of Faner et al. (2019) and Cardarelli et al. (2002). Actual meal particles are irregular; the radius selected for each experiment is therefore stated with its source and interpretation. Fixed radius and porosity separate transport from swelling and shrinkage, for which no constitutive laws are supplied here. The particle's total n-hexane loading X_h (every loading here in kg per kg of dry, water- and solvent-free meal) is compared with two reference loadings, both for hexane rather than water. The critical hexane loading $X_\mathrm{c}(T)$ marks exhaustion of external free hexane and the constant-rate to falling-rate transition, analogous to critical moisture in drying terminology. It is a capacity reference, anchored to the transition measured by Faner et al. (2019) (Table 2), not an equilibrium isotherm. The equilibrium hexane reference $X^*_\mathrm{e}(T, P)$ is the retained loading evaluated at the pressure-feasible saturated activity a^* of Eq. (1). It does not prescribe the loading at core extinction. In regime A, external free hexane wets the outer surface at unit hexane activity and evaporation is supplied through the external heat-transfer boundary. In regime B ($0 < s < R_\mathrm{p}$), liquid remains behind a receding interface. The chosen total-vapour closure makes its recession heat-limited; binary molecular diffusion in the shell remains active, as explained below. This is a model assumption about the connected vapour path, not a general claim that molecular diffusion cannot limit industrial desolventization. In regime C, no liquid-hexane core remains: $s = 0$. Retained material and pore gas continue to obey the same local component and energy balances. Core extinction is a geometric conservation event; a particle-mean loading is not used to prescribe it. The saturated equilibrium loading is a thermodynamic reference rather than a uniform-particle transition rule (Figure 1).

Figure 1 follows external free-hexane depletion ($X_\mathrm{f} = 0$) and subsequent core disappearance ($s = 0$). Purple represents hexane and blue represents water; orange arrows show incoming heat. The dashed blue band is the optional external water film, through which hexane can escape in the proposed closure. Liquid-water imbibition operates in the dry shell connected to that film, with

Table 1: Governing balances and supporting closures of the particle model. Compact expressions omit repeated state arguments; the cited equations give the full definitions.

Block / equations	Compact expression	Physical role	Source / derivation
Conservation (2)–(3)	$\partial_t C_k + \mathcal{D}_r(j_k) = 0$ $\partial_t u + \mathcal{D}_r(J_E) = 0$ $J_E = q + \sum_k \bar{h}_k j_k$	Evolve water, hexane and energy with no separate phase-change source.	Component and energy conservation; regional coupling developed here.
Inventories (4)–(6)	$C_k^{\mathrm{d}} = \varepsilon_g c y_k + \rho_{\mathrm{dm,p}} W_k / M_k$ $\mathrm{D}_s(\rho_{\mathrm{dm,p}} X^a_{\mathrm{h,w}})/\mathrm{D}t = 0$	Store pore gas and retained material separately; preserve the wet-core donor loading.	Present regional storage and historical-core closure.
Internal transport (7)–(9)	$J_{\mathrm{w}} = -J_{\mathrm{h}} = -L^{\mathrm{eff}}_{\mathrm{wh}} \nabla_{\mathrm{comp}} \Xi$ $\Xi = (\mu_{\mathrm{w}} - \mu_{\mathrm{h}})/(RT)$ $N_i^{\mathrm{g}} = y_i N_t + J_i$	Couple binary gas exchange to a separate retained-water path; obtain N_t from continuity.	Maxwell–Stefan framework (Krishna & Wesselingh, 1997); present force reconstruction.
Moving front (13)–(17)	$[j_k - \dot{s} C_k]^{\mathrm{d}}_{\mathrm{w}} = 0$ $[J_E - \dot{s} u]^{\mathrm{d}}_{\mathrm{w}} = 0$	Determine front recession from component and energy jumps, using the same swept donor.	Moving-boundary conservation (Crank, 1984); present donor-restricted closure.
Surface stocks (20)–(22)	$\dot{L}_{\mathrm{w}} = j_{\mathrm{w},p} - j_{\mathrm{w},g}$ $\mathrm{d}(L_{\mathrm{w}} u_{\mathrm{w},l})/\mathrm{d}t = J_{E,\mathrm{p}} - J_{E,\mathrm{g}}$	Conserve external condensate and its energy, with hexane continuity through the film.	Present surface balances; hexane-transparent water film is an assumption.
Gas-film exchange (23)–(25)	$q_{\mathrm{in}} = h_Q \Phi(\beta)(T_b - T_s)$ $\Phi(x) = x/(e^x - 1)$	Couple finite-flow component exchange, heat supply and transported enthalpy.	Analytic film closure adopted here; bed coefficient from Faner (2008).
Water imbibition (26)–(27)	$G_\ell = (d/\mathcal{L}_\ell + 1/K_\ell)^{-1}$ $N_\ell = -G_\ell \Delta \ln a_{\mathrm{w,ret}}$	Add finite liquid-connected entry, limited by available external water.	Present effective series closure; coefficient construction uses Hemmingsen et al. (2008).
Retention (28)–(30)	$W_{\mathrm{h}} = \alpha_o W_{\mathrm{h,native}} + \beta_o q_o$ $W_{\mathrm{w}} = A_1/[1 + A_2 \ln(1/a_{\mathrm{w}})]$	Close component storage and activities; adjust oil content relative to reference meal.	GAB (Cardarelli & Crapiste, 1996); water fit (Luz, Sousa, Jorge, & Paraíso, 2006); present oil anchoring.
Property laws (12), (31)–(32)	$D_b \propto T^{1.75}/P$ $c_{p,\mathrm{o}} = 1024 + 3T$ $\Delta u_{\mathrm{dm}} = \int c_{p,\mathrm{dm}}\, \mathrm{d}T$	Evaluate pore mobility and dry-matrix sensible energy at the evolving temperature.	Shape: Fuller, Schettler, and Giddings (1966), assumed pore magnitude; oil: Garcia Rojas, Coimbra, and Telis-Romero (2013).

$\mathcal{D}_r(F) = r^{-2}\partial_r(r^2 F)$; s is the core radius. Local j_k is molar, while surface j_k is a mass flux ($j_k^{\mathrm{mass}} = M_k j_k^{\mathrm{molar}}$). Fluxes are outward-positive. T is in kelvin and $c_{p,\mathrm{o}}$ in J/(kg K). The compact map does not replace the phase, capacity and caloric-consistency conditions in Eqs. (18), (19) and (34). Detailed property sources and the separate historical Fickian comparator are in Sec. S14.

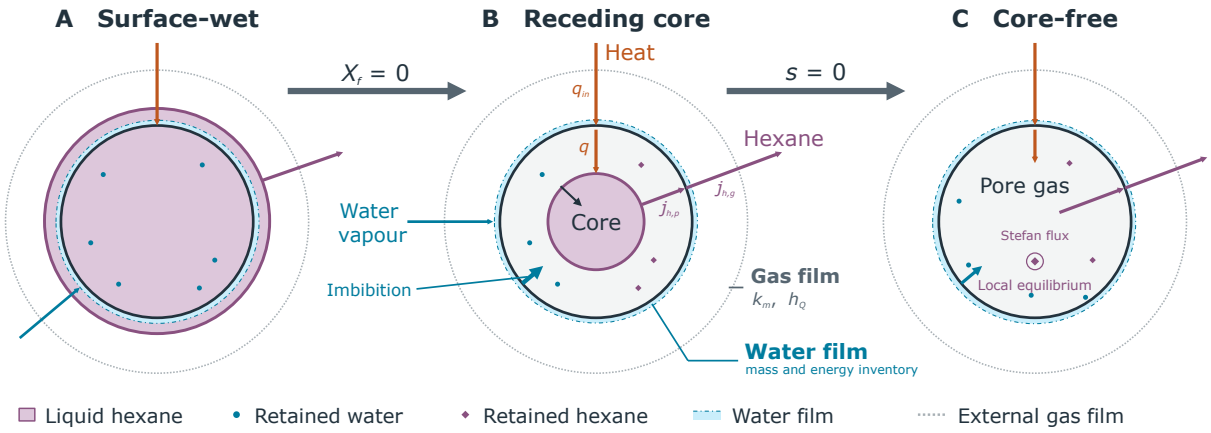


Figure 1: Particle regimes and transport pathways. X_f: external free-hexane loading; s: liquid-hexane core radius; R_p: particle radius.

no additional liquid-water flux into the liquid-hexane core. Dots and diamonds represent retained water and retained hexane, respectively. Geometry and film thickness are schematic. In this figure and the surface-film balances, j is a mass flux and N a molar flux, related by $j_k^{\mathrm{mass}} = M_k N_k$; $h_k^{\mathrm{mass}} = \bar{h}_k / M_k$, with M_k the component molar mass. The local balances below instead use molar j_k. All fluxes are outward-positive; subscripts p and g identify the particle and gas sides of the surface.

The particle model is written so that it can later be coupled to a tray model. For prescribed gas and heat-transfer conditions, the particle balances return outward-positive surface fluxes $j_\mathrm{w}(R_\mathrm{p})$, $j_\mathrm{h}(R_\mathrm{p})$ and $J_E(R_\mathrm{p})$ on one caloric datum. These are responses to the evolving interior, rather than imposed drying rates. For particle classes c with total exposed areas $A_{\mathrm{ex},c}$, the rates delivered to a future tray gas balance would be $\sum_c A_{\mathrm{ex},c} j_{k,c}(R_{p,c})$ for component k and $\sum_c A_{\mathrm{ex},c} J_{E,c}(R_{p,c})$ for total energy. Condensation has a negative water rate; each gas source has an equal and opposite entry in the particle balance. The tray would return its updated gas state to the particles. In this paper, the gas and bed conditions along the equipment trays are prescribed instead. This separates the particle response from countercurrent gas mixing and tray residence, which is useful for testing the particle equations before coupling them to DT design calculations; steam consumption itself requires those equipment balances (Kemper, 2013).

The equilibrium boundary is evaluated at the *pressure-feasible* hexane activity

$$a^*(T,P) \;=\; \min\!\Big(1,\; P/P_{\mathrm{sat}}^{\mathrm{h}}(T)\Big)\,, \tag{1}$$

with P the total pressure and $P_{\mathrm{sat}}^{\mathrm{h}}$ the n-hexane vapour pressure, both in pascals. Equation (1) is an ideal-mixture diagnostic. The pore and interface equilibrium used in the calculations uses binary-virial gas fugacities and the pure-liquid equation-of-state fugacities on the same pressure and caloric datum. It does not omit those corrections when locating phase boundaries. Above the solvent's boiling point at the bed pressure, an equilibrium loading taken at unit activity is overstated. Evaluated with (1), the two boundaries stay one to two orders of magnitude apart and separate further as the temperature rises; the paper uses only that *ordering* (Sec. S6).

The external free-hexane loading is $X_\mathrm{f} = m_{\mathrm{h,ext}}/m_{\mathrm{dm}}$: a finite mass of external free liquid hexane assigned to the particle, per unit dry meal mass. It is neither a liquid composition nor a gas concentration. It can decrease continuously while the remaining pure liquid still has unit hexane activity. Its mass and energy enter the regime-A balances, although its spatial thickness

and liquid flow are not resolved. The partition between external and intraparticle hexane follows the critical-volume capacity at the current liquid density; their sum changes through the surface hexane flux. The A–B transition occurs when this inventory is exhausted. While $X_\mathrm{f} > 0$, the front stays at $s = R_\mathrm{p}$; once $X_\mathrm{f} = 0$, it may recede and leave a dry shell. Hexane vapour can still be present around the particle when $X_\mathrm{f} = 0$. This sequential-drainage condition is stated in Eq. (18). The reference critical loading of 0.20 kg/kg measured by Faner et al. (2019) sets the effective critical-volume capacity at the reference liquid density, 615 kg/m^3. At the particle's actual temperature, $X_\mathrm{c}(T) = \kappa_v \varepsilon_\mathrm{g} \rho_\mathrm{h,l}(T,P)/\rho_\mathrm{dm,p}$, with $\kappa_v = 2.675$ fixed by that reference capacity. Here $\rho_\mathrm{h,l}(T,P)$ is the liquid n-hexane density from the technical Helmholtz equation of state of Span and Wagner (2003a, 2003b), also used for its liquid energy and fugacity. Water uses IAPWS-95 (IAPWS, 2018; Wagner & Pruß, 2002); the binary pore gas uses a second-virial mixture with a water–hexane cross term (Plyasunov & Shock, 2003) (Section 2.8). The critical hexane loading therefore varies with liquid density, while the time at which it is reached follows from the regime-A inventory balance under the prescribed gas and heat history. The component and energy jumps then determine the recession speed and the dry-side hexane flux at the front. The shell balances and surface conditions determine the corresponding particle-surface exchanges. In the discrete model the front first departs over an accepted time step, giving a finite shell thickness without assigning the shell the width of one radial cell (Sec. S4). The A–B boundary is thus a surface-state constraint, with no fitted time or drying-rate selector. The stationary binary regime-C closure starts after the liquid-hexane core has vanished. In a calculation, the crossing *loading* therefore reproduces a supplied value; the crossing *time* is the tested prediction, conditional on the external history (Section 4.1).

Heat crosses the gas film and dry shell in series. The external thermal resistance dominates until the core radius falls to about $0.28R_\mathrm{p}$; the shell dominates later (Sec. S1). The heat-fed front closure and its omitted total-vapour resistance are explained in Section 2.3. The 2019 and older Faner records motivate different transport descriptions; the older records use the separate Fickian comparison of Eq. (S2), as discussed in Sec. S3.

Temperature and composition also vary over different radial scales. The earlier constant-property dry-shell Lewis number of 223.25 illustrates why their profiles must be reconstructed separately near the moving front (Crank, 1956; Perré, Rémond, Almeida, Augusto, & Turner, 2023).

2.2 Local balances and region closures

Inside each region the model is a set of conservation laws with no source terms, closed differently in the liquid-hexane core and in the dry shell. The sphere holds two conserved mobile components, water and n-hexane, and an immobile composite dry meal of local density $\rho_\mathrm{dm,p}$ with a residual-oil mass fraction w_o. A sharp interface at radius $r = s(t)$, with t the time, separates a liquid-hexane core (w) from a dry shell (d); the region label $\alpha \in \{\mathrm{w}, \mathrm{d}\}$ identifies these physical regions, not a numerical cell. The component index $k = \mathrm{w}$ means water, whereas the region superscript w means the liquid-hexane core. Radial fluxes are positive outward, and recession means $\dot{s} < 0$. The total radial energy flux is $J_E^\alpha = q^\alpha + \sum_k \bar{h}_k^\alpha j_k^\alpha$, with q in W/m^2, component molar flux j_k in mol/(m^2 s) and partial molar enthalpy $\bar{h}_k$ in J/mol. A partial molar enthalpy is the change in total enthalpy when a mole of that component is added at fixed temperature, pressure and amounts of the other components. For each transported stream it is evaluated at the local state, on the common datum of Section 2.7. The balances below hold inside either region, away from the sharp front, where the

fields are differentiable. The front itself requires the jump conditions of Section 2.3.

$$\frac{\partial C_k^\alpha}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2 j_k^\alpha\Big) = 0, \qquad k \in \{\mathrm{w}, \mathrm{h}\}, \tag{2}$$

$$\frac{\partial u^\alpha}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2 J_E^\alpha\Big) = 0, \qquad j_{\mathrm{dm}} = 0, \qquad j_{L_\mathrm{o}} = 0, \tag{3}$$

Here C_k^α is the total amount of component k in all its local forms per unit particle volume (mol/m^3), not its concentration in one phase. For example, in a gas-filled dry shell,

$$C_k^\mathrm{d} = \varepsilon_\mathrm{g} c\, y_k + \rho_{\mathrm{dm,p}} W_k / M_k, \tag{4}$$

where c is pore-gas molar density, y_k its mole fraction, ε_g gas-filled porosity, and W_k retained loading in kg/kg dry. Phase-specific concentrations need not be equal; local equilibrium relates their chemical potentials rather than their concentrations. The internal energy density u likewise sums stored contributions, including the dry meal. The immobile dry meal and residual oil have zero fluxes j_{dm} and j_{L_o}. Retention, evaporation and condensation redistribute a component between local phases and are therefore *not* sources in (2): no phase-change term appears in the smooth-region equations (Sec. S1).

Two statements close the liquid-hexane core. First, no relative n-hexane flux is carried inside the mobile-liquid region: at activation, every material shell stores the historical label $X^a_{\mathrm{h,w}} = X_\mathrm{c}(\theta_a)$, the critical loading at its temperature at activation θ_a, and thereafter

$$\frac{\mathrm{D}_s}{\mathrm{D}t}\Big(\rho_{\mathrm{dm,p}}\, X^a_{\mathrm{h,w}}\Big) = 0 \tag{5}$$

following the solid. Thus $X^a_{\mathrm{h,w}}$ is the apparent mobile-liquid hexane loading of the liquid-hexane core, stored at activation and distinct both from the particle's total loading X_h and from the retained hexane of the dry shell. The core is liquid-filled, the hexane activity is unity, and the non-negativity of the mobile liquid is an admissibility condition, not a clipped quantity. Second, the core is resolved, not lumped, with specific energy per kilogram of dry meal

$$e_\mathrm{w} \;=\; h_\mathrm{dm}(T, w_\mathrm{o}) \;+\; X_\mathrm{w}\, h_\mathrm{w,l}(T,P) \;+\; X^a_{\mathrm{h,w}}\, u_\mathrm{h,l}(T,P) \;-\; B_\mathrm{h}(a_\mathrm{h}{=}1, T, w_\mathrm{o}), \tag{6}$$

with h_dm the specific enthalpy of the dry meal, X_w the total water loading, and $h_\mathrm{w,l}$ and $u_\mathrm{h,l}$ the specific enthalpy of liquid water and the specific internal energy of liquid n-hexane, both in J/kg. Retained and mobile hexane are not carried separately in this energy, so the binding potential B_h is counted once and the repartition between them is source-free.

The dry shell stores each component in two forms, pore gas and a retained contribution, with distinct storage terms and distinct thermodynamics; no retained-only symbol includes pore gas. Transport consists of one thermodynamic-force pore flux, one retained-matrix water flux and one algebraic total flux,

$$J_\mathrm{w} \;=\; -J_\mathrm{h} \;=\; -L^\mathrm{eff}_\mathrm{wh}\, \nabla_\mathrm{comp}\left[\frac{\mu_\mathrm{w} - \mu_\mathrm{h}}{RT}\right], \tag{7}$$

$$N_\mathrm{w,ret} \;=\; -L^\mathrm{eff}_\mathrm{w,ret}\, \nabla_\mathrm{comp}\left[\frac{\mu_\mathrm{w,ret}}{RT}\right], \qquad N_i^\mathrm{g} = y_i N_\mathrm{t} + J_i, \tag{8}$$

with μ_w and μ_h the pore-gas chemical potentials of water and n-hexane and $\mu_\mathrm{w,ret}$ that of retained water (J/mol), R the gas constant, $L^\mathrm{eff}_\mathrm{wh}$ and $L^\mathrm{eff}_\mathrm{w,ret}$ the pore and retained-matrix mobilities (mol/(m s)), y_i the pore-gas mole fraction of component i, and the fluxes J_i, $N_\mathrm{w,ret}$, N_i^g and N_t molar

(mol/(m^2 s)). The total (Stefan) flux N_t is recovered algebraically from the summed continuity equations rather than fitted. It rests on an Onsager mobility whose degeneracy vanishes exactly at either purity and needs no composition floor (Krishna & Wesselingh, 1997). The shell energy flux carries each stream exactly once, with no latent, binding, pressure-volume or Stefan-energy correction on top (Sec. S1.1). We retain composition-driven mass transport and conductive heat transport, and neglect independent Soret terms because no thermal-diffusion coefficients for this meal are supplied. This is a constitutive approximation; it does not show that thermal diffusion is negligible. Here ∇_comp denotes the composition part of the fixed-(T,P) chemical force, not the full gradient of a temperature-dependent standard chemical potential. At ordinary faces, the isothermal composition secant is evaluated at a common face temperature. The moving-interface reconstruction integrates that composition derivative along the actual linear temperature/composition path,

$$G_\mathrm{comp}(\Xi) = \frac{1}{d}\int_0^1 \left.\frac{\partial \Xi}{\partial y_\mathrm{h}}\right|_{T(\lambda),P} \frac{\mathrm{d}y_\mathrm{h}}{\mathrm{d}\lambda}\,\mathrm{d}\lambda, \qquad \Xi = \ln(f_\mathrm{w}/f_\mathrm{h}) \text{ or } \ln(f_\mathrm{w}/f_\mathrm{ref}). \tag{9}$$

Here d is the face separation and f_i is the equation-of-state fugacity. Energy transport remains non-isothermal and includes component enthalpies. This is the declared nominal transport approximation in both prescribed-bath experiments. The pressure variation omitted by taking N_t from continuity is estimated from Darcy's law across the dry shell,

$$\Delta P \;\approx\; \frac{N_\mathrm{t}\,RT}{P}\,\frac{\mu_g}{B}\,(R_\mathrm{p} - s), \tag{10}$$

with μ_g the pore-gas viscosity and B the Darcy permeability. For an illustrative vapour-flux bound, take $|N_\mathrm{t}| = 0.5\,\mathrm{mol/(m^2\,s)}$; this is a supplied sensitivity scale, not the measured or calculated mean flux of the new DT history. With this flux bound, $\mu_g = 10^{-5}\,\mathrm{Pa\,s}$, T at the heteroazeotrope and a shell 1 mm thick, the estimate gives ΔP of about 0.15 kPa, 1.5 kPa and 15 kPa at $B = 10^{-12}$, 10^{-13} and $10^{-14}\,\mathrm{m^2}$. A capillary bundle $B = \varepsilon_g r_\mathrm{pore}^2/(8\tau)$ at the particle porosity 0.14 and a tortuosity of three is about $10^{-13}\,\mathrm{m^2}$ for a 5 ţm pore (ΔP about 1 kPa on the same bound) and about $6\times10^{-15}\,\mathrm{m^2}$ for a 1 ţm pore (ΔP about 25 kPa). Below about $10^{-15}\,\mathrm{m^2}$ the estimate exceeds the bed pressure. The isobaric closure therefore assumes that the vapour crosses a connected pore space of several micrometers. It does not apply to a path confined to sub-micrometer pores, and no measured permeability of this meal is supplied here. Molecular diffusion is already present in Eq. (7): J_k is the exchange of the two vapours relative to their molar-average motion. With $L_\mathrm{wh}^\mathrm{eff} = c\,y_\mathrm{w}y_\mathrm{h}D_b$, the ideal-gas, fixed-$(T,P)$ composition force gives

$$J_\mathrm{h} = -cD_b\frac{\partial y_\mathrm{h}}{\partial r}, \qquad N_\mathrm{h}^\mathrm{g} = y_\mathrm{h}N_\mathrm{t} - cD_b\frac{\partial y_\mathrm{h}}{\partial r}. \tag{11}$$

Here D_b is the effective binary pore diffusivity. The first term in N_h^g carries hexane with the total vapour motion; the second separates hexane and water by molecular diffusion. The calculations retain the non-ideal chemical force of Eq. (7), rather than imposing the ideal-gas identity of Eq. (11). The pore coefficient is evaluated at the local thermodynamic-force temperature and pressure:

$$D_b(T,P) = D_{b,\mathrm{ref}}\left(\frac{T}{378.15\,\mathrm{K}}\right)^{1.75}\frac{101325\,\mathrm{Pa}}{P}, \qquad D_{b,\mathrm{ref}} = 4.0\times10^{-10}\ \mathrm{m^2/s}. \tag{12}$$

The $T^{1.75}/P$ shape follows Fuller et al. (1966), assuming constant material hindrance when transferred to an effective pore coefficient. The reference magnitude is an unvalidated engineering assumption guided by the scalar hexane value 4.0×10^{-10} m^2/s at 105 °C used by Cardarelli et al. (2002) from rapeseed evidence. That source's scalar flux and explicit porosity differ from this molar

binary closure; no coefficient conversion has been established. Porosity is already included in D_b. Reference magnitudes 2, 4 and 6×10^{-10} m^2/s define a prescribed sensitivity range, not measured uncertainty (Sec. S13). Summing Eq. (8) gives $N_{\mathrm{w}}^{\mathrm{g}} + N_{\mathrm{h}}^{\mathrm{g}} = N_{\mathrm{t}}$, recovered from continuity. Only when $N_{\mathrm{t}} = 0$ does the total hexane flux equal the diffusive exchange flux; writing it as a gradient of hexane concentration also requires constant total gas concentration. At pure n-hexane, binary exchange vanishes but the outward flux can remain N_{t} (Secs. S1.1 and S1.1). The scalar Fick description used for the older drying records is a separate comparison with an effective transport coefficient, given in Eq. (S2). In both prescribed baths, regime C instead integrates the stationary radial binary component and total-energy balances. Its initial inventories and temperatures come from the conservative core-extinction projection. The same particle equations therefore carry solvent removal through core disappearance and the remaining drying tail.

Sharing the component and energy balances couples solvent removal to the heat carried out, particle temperature and water condensation or desorption. Retained water has its own transport path, and retained hexane contributes storage and binding energy. These effects are solved together rather than added to a hexane-only result. The separate historical Fickian comparison does not establish an additional branch of this binary particle law.

2.3 Front conditions

The front conditions fix the front velocity from the heat that reaches the front. For a spherical control volume with moving inner and outer radii $r_1(t)$ and $r_2(t)$, conservation of either component and of total energy requires

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(4\pi \int_{r_1}^{r_2} C_k r^2 \,\mathrm{d}r\right) = -4\pi r_2^2 (j_k - \dot{r}_2 C_k)_{r_2} + 4\pi r_1^2 (j_k - \dot{r}_1 C_k)_{r_1}, \qquad k \in \{\mathrm{w}, \mathrm{h}\}, \tag{13}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(4\pi \int_{r_1}^{r_2} u r^2 \,\mathrm{d}r\right) = -4\pi r_2^2 (J_E - \dot{r}_2 u)_{r_2} + 4\pi r_1^2 (J_E - \dot{r}_1 u)_{r_1}. \tag{14}$$

The terms proportional to $\dot{r}_1$ and $\dot{r}_2$ account for material swept through the moving boundaries. Shrinking this volume onto $r = s(t)$ gives the Rankine–Hugoniot conditions for both components and total energy (Crank, 1984),

$$[\![\, j_k - \dot{s}\, C_k \,]\!]_{\mathrm{w}}^{\mathrm{d}} = 0, \qquad k \in \{\mathrm{w}, \mathrm{h}\}, \tag{15}$$

$$\left[\!\!\left[\, q + \textstyle\sum_k \bar{h}_k j_k - \dot{s}\, u \,\right]\!\!\right]_{\mathrm{w}}^{\mathrm{d}} = 0, \tag{16}$$

with $[\![\cdot]\!]_{\mathrm{w}}^{\mathrm{d}}$ the dry-side value minus the wet-side value. Two different wet-side states occur in these conditions and must be kept apart. The first is the equilibrium interface trace, which the interface conditions solve for. The second is the *donor* state, the one-sided limit of the wet-core fields themselves. Every term multiplying $\dot{s}$ is a stored state, swept at the state of its own side. With the zero wet-side relative hexane flux of (5), the hexane jump fixes the dry-side flux that the moving front produces, $j_{\mathrm{h,d}} = \dot{s}(C_{\mathrm{h,d}}^{\Gamma} - C_{\mathrm{h,w}}^{\mathrm{b}})$. The wet-side donor value, the historical label stored at activation rather than a re-solved critical loading, must exceed the dry-side trace. Substituting this flux into (16) moves the term $\bar{h}_{\mathrm{h,d}} j_{\mathrm{h,d}}$, which is proportional to $\dot{s}$, into the denominator. The front velocity is then

$$\dot{s} \;=\; -\,\frac{[\![q]\!]_{\mathrm{w}}^{\mathrm{d}} + \bar{h}_{\mathrm{w,d}}\, j_{\mathrm{w,d}} - \bar{h}_{\mathrm{w,w}}\, j_{\mathrm{w,w}}}{\bar{h}_{\mathrm{h,d}}(C_{\mathrm{h,d}}^{\Gamma} - C_{\mathrm{h,w}}^{\mathrm{b}}) - (u_{\mathrm{d}}^{\Gamma} - u_{\mathrm{w}}^{\mathrm{b}})}, \tag{17}$$

a heat-supply to stored-state-jump quotient in its donor-restricted form. The numerator is the conduction jump, dry side minus wet side, plus the enthalpy carried by the water flux. That water

flux is evaluated from the water jump and the shell. The denominator is the energy required to sweep a volume of liquid-hexane core into the dry-side state. On the common datum of Section 2.7 it contains the latent heat of the hexane removed, a smaller ideal-gas correction and the binding energy already stored in u (Sec. S1). The superscript Γ marks the equilibrium interface trace, b the donor state, and $\bar{h}_{\mathrm{h,d}}$ the partial enthalpy of n-hexane on the dry side. Binary exchange separates the vapours but does not throttle their algebraic total flux. Recession is therefore heat-fed, with no Darcy or Knudsen resistance on total vapour motion; shell Péclet numbers inside the reported band are of order 10^3 (Sec. S6). A transport-controlled front would require an additional total-flux law. The separate Fickian comparator is not applied here, and no empirical rate selector is used. After core extinction, regime C retains the same binary radial closure.

2.4 Admissible states and the removal of swept material

The front conditions hold in the discrete model only if states outside the admissible set are rejected and swept material is removed at one consistent state. The interface is closed by continuity of temperature, local phase equilibrium at unit hexane activity on the wet side, and equal liquid and gas pressure; the last condition means that macroscopic capillarity is omitted. Admissibility is a complementarity system rather than a clipping rule,

$$X_{\mathrm{f}} \geq 0, \qquad R_{\mathrm{p}} - s \geq 0, \qquad X_{\mathrm{f}}\,(R_{\mathrm{p}} - s) = 0, \qquad \dot{s} \leq 0, \tag{18}$$

whose first three conditions express strict sequential drainage (the external free-hexane loading X_{f} must be exhausted before the front may leave the outer surface) and whose fourth restricts the formulation to primary drainage. A trajectory that violates any of them, inverts the interfacial concentration jump or loses interior mobile liquid is rejected at that step. The violated bound is reported and the last accepted state is left unchanged: no second interface is created and no state is repaired to keep the integration running.

One consistency requirement needs a separate statement, because a discretization can violate it silently. When the interface sweeps material, the concentration and the energy density used to debit the swept volume must be read from the same material state. Using the equilibrium trace for one and the consumed material's bulk state for the other would imply a pre-drainage that the intraparticle Péclet number of this system excludes. Both are therefore taken at the bulk, old-time state of the consumed piece. In that state the retained water is *continuous* across the front, because hexane drains and water does not (Sec. S1.1). The formal statement is a gauge covariance: shifting the arbitrary caloric zero of water by a constant c_{w} must move the discrete energy residual of the interface rows by exactly c_{w} times their discrete water residual,

$$\frac{\partial R_E}{\partial c_{\mathrm{w}}} \;=\; R_{\mathrm{w}}, \tag{19}$$

identically in the discrete state. Section 3.4 uses this identity as the acceptance test for the interface treatment. The same test is available to any moving-boundary phase-change discretization: if the energy residual does not move exactly with the component residual when that component's caloric zero is shifted, latent or binding heat is being counted twice, or the swept material is being debited from the wrong state.

2.5 The conserved external water film

When the bath supplies condensing water, the model stores it in an external water film, an inventory separate from retained water. Steam condensation supplies heat for desolventization while increasing

the water that may require removal in the downstream dryer (Kemper, 2013; Witte, 1995). Keeping the film separate lets the balances determine how much water enters the particle and how much evaporates back to the bath. We use one film temperature and inventory at the surface, rather than resolving its liquid flow or spatial thickness. Let L_{w} be its mass per unit particle area, $j_{k,\mathrm{p}}$ the outward pore-side component flux, and $j_{k,\mathrm{g}}$ the outward gas-film flux. To isolate the effect of water uptake, the external water film is assumed to add no resistance to hexane escape. This limiting case is not a measured water-film permeability; a blocking effect would require an additional constitutive law. The surface balances are

$$\frac{\mathrm{d}L_{\mathrm{w}}}{\mathrm{d}t} = j_{\mathrm{w,p}} - j_{\mathrm{w,g}}, \tag{20}$$

$$0 = j_{\mathrm{h,p}} - j_{\mathrm{h,g}}, \tag{21}$$

$$\frac{\mathrm{d}[L_{\mathrm{w}} u_{\mathrm{w},l}(T_{\mathrm{s}})]}{\mathrm{d}t} = J_{E,\mathrm{p}} - J_{E,\mathrm{g}}. \tag{22}$$

The gas-film closure uses an ideal binary mass-coordinate partition, with equation-of-state phase traces and partial enthalpies. For total outward mass flux j_t, conductance $K_f = \rho_f k_m$, and hexane mass fractions $Y_{\mathrm{h},s}$ and $Y_{\mathrm{h},b}$,

$$j_{\mathrm{h},g} = Y_{\mathrm{h},s} j_t + K_f (Y_{\mathrm{h},s} - Y_{\mathrm{h},b}) \Phi(j_t/K_f), \qquad j_{\mathrm{w},g} = j_t - j_{\mathrm{h},g}, \tag{23}$$

$$\Phi(x) = \frac{x}{e^x - 1}, \quad \Phi(0) = 1, \qquad \beta = \frac{j_{\mathrm{w},g} c^f_{p,\mathrm{w}} + j_{\mathrm{h},g} c^f_{p,\mathrm{h}}}{h_Q}, \tag{24}$$

$$q_{\mathrm{in}} = h_Q \Phi(\beta)(T_{\mathrm{b}} - T_{\mathrm{s}}), \qquad J_{E,\mathrm{g}} = -q_{\mathrm{in}} + \sum_k \frac{j_{k,\mathrm{g}}}{M_k} \bar{h}_k(T_{\mathrm{s}}). \tag{25}$$

The component heat capacities are component partial-enthalpy secants between surface and bulk temperatures. Both component capacity fluxes enter β, including counter-diffusion. The same analytic Ackermann expression is used within a declared band $|\beta| \leq 3$. Its largest accepted value in the recorded calculations is 1.2063, above the previously assessed film band $|\beta| \leq 1$. This wider band is a declared assumption about the film law, separate from the numerical root tolerances; it has not been demonstrated for this gas film. All advected partial enthalpies enter the two total-energy fluxes once. Positive external liquid selects unit surface water activity; with zero liquid, gas-side composition is solved from the same three balances and may remain undersaturated. Appearance and depletion are conservative events, not resets of the film mass or temperature. The interior retention capacity is a separate active set. The numerical unit-activity trace of the hexane-transparent film and its sensitivity are stated with the experiments. For the regime-A surface in the pure-hexane bath, the hexane-only surface active set sets the outward water flux to zero while external free hexane remains. The water in the particle stays stored and becomes exposed through the binary pore gas once the external free hexane is depleted.

2.6 Liquid-water imbibition from the external water film

Liquid-water imbibition is assumed to occur when the external water film contacts the dry shell. Steam condenses on the particle faster than the water can move inward, so a liquid film accumulates on the surface and is drawn into the pore structure afterwards, where the matrix and the interstitial space retain it (Karnofsky, 1985). Liquid contact also admits water that vapour sorption does not: the imbibition capacity of soy protein isolates exceeds their vapour-sorption capacity several-fold (Jovanovich et al., 2003). Those measurements concern isolates at 20 °C, rather than hot extracted meal. We represent this path as an effective finite-conductance pathway into the existing

capacity-bounded retained-water inventory. The closure does not identify every absorbed molecule as strongly bound and adds no independently resolved pore-liquid domain. Vapour transport and retention remain active through the original component equations.

Let $\psi = \ln a_{\mathrm{w,ret}}$ be the activity potential evaluated from the actual retained-water state. For the additional liquid-water mobility $\mathcal{L}_\ell > 0$, liquid contact conductance $K_\ell > 0$ and distance $d > 0$ between the outer cell and surface trace, the external contribution is

$$G_\ell = \left(\frac{d}{\mathcal{L}_\ell} + \frac{1}{K_\ell}\right)^{-1}, \qquad N_\ell = -G_\ell(\psi_{\mathrm{s}} - \psi_{\mathrm{p}}). \tag{26}$$

N_ℓ is outward-positive in mol/(m^2 s). Thus $0 < G_\ell \le K_\ell$ and $G_\ell \to K_\ell$ as $d \to 0$: when the dry shell first forms, the surface conductance stays finite. Interior faces in the dry shell use $G_\ell = \mathcal{L}_\ell/d$. This added path is absent at the remaining wet-core interface and when external liquid is absent. Surface and cell activities follow their capacity-bounded retained states; the added force is not obtained by substituting unit pure-liquid activity for the retained surface trace.

For available film stock $L_{\mathrm{w},old}$, inward addition is limited by

$$j_{\ell,\mathrm{in}} \le \max\left(0, \frac{L_{\mathrm{w},old}}{\Delta t} + j_{\mathrm{w,p}}^{\mathrm{base}} - j_{\mathrm{w,g}}\right). \tag{27}$$

The final flux enters $j_{\mathrm{w,p}}$ in Eq. (20) and the particle balance with opposite signs. Its donor retained-water partial enthalpy enters the energy flux once. Under the negligible-Soret approximation for this additional path, its composition-force dissipation is proportional to $RG_\ell(\psi_{\mathrm{s}} - \psi_{\mathrm{p}})^2 \ge 0$. Liquid-water imbibition is an effective transport hypothesis, not a resolved capillary-pressure, hydraulic saturation or swelling model. Water displacement of pore hexane is neglected, and the film adds no resistance to hexane escape (Section 2.5).

We estimate the total liquid-connected mobility as $100L_{\mathrm{w,ret}}^{\mathrm{eff}}$, so the added mobility is $\mathcal{L}_\ell = 99L_{\mathrm{w,ret}}^{\mathrm{eff}}$. The anchor is the mean 1.43-mm soybean-meal size and 90-s hot-water uptake scale reported by Hemmingsen et al. (2008). An assumed spherical internal-diffusion interpretation, transferred from 80 to 100 °C by inverse water viscosity, gives an apparent diffusivity of about (1.8–3.0) $\times 10^{-9}$ m^2/s. At $W = 0.20$ kg/kg dry the baseline retained-water path gives 2.4×10^{-11} m^2/s, hence an estimated factor of 76–125, rounded to 100. The construction and its cake-water, geometry and endpoint assumptions are given in Sec. S12. The finite contact conductance is $K_\ell = 0.32$ mol/(m^2 s). The two coefficients are not jointly fitted to the source uptake curve. Neither coefficient is fitted to DT outlet measurements.

2.7 One caloric datum per component

All phases of a component share one caloric zero. Latent and binding effects enter through stored-state differences and component enthalpy fluxes, so an additional vaporization or fitted sorption-heat source would double-count energy. Shifting the common zero must leave the solution unchanged, as tested in Section 3.4. For hexane, the sorption potential and binding potential B_{h} form the Gibbs–Helmholtz pair derived from its temperature-dependent isotherm (Sec. S1.1). The gas-stream enthalpies are evaluated as $\bar{h}_{k,\mathrm{g}} = h_{k,\mathrm{g}}^{\mathrm{ideal}}(T) + \bar{h}_{k,\mathrm{g}}^{\mathrm{res}}(T, P, y_{\mathrm{w}}, y_{\mathrm{h}})$. The ideal terms follow the pure-component property formulations; the residual partial molar terms follow the binary virial mixture. Pure-fluid property sources are given in Section 2.8. The nominal retained-water energy uses liquid-water enthalpy on the same datum. Its temperature-independent Luikov activity law contributes no separate excess water-binding heat. Condensation and evaporation energy are already present through phase-state differences and component fluxes.

2.8 Sorption isotherms, property equations and parameters

The sorption isotherms and property equations close the model, and the ranges over which they were measured set where its results can be trusted. Retained n-hexane on reference soybean meal follows the Guggenheim–Anderson–de Boer isotherm fitted by Cardarelli and Crapiste (1996),

$$W_{\text{native}}(a_{\text{h}}, T) = \frac{X_m\, C\, x}{(1-x)\,(1-x+C\,x)}, \qquad x = K(T)\, a_{\text{h}}, \tag{28}$$

with X_m the monolayer loading and C and K the dimensionless GAB constants. Both C and K have an Arrhenius temperature dependence, and no activity clamp is applied. The sorbing sample carried reference oil, so (28) is a *total-meal* isotherm, with no duplicate oil inventory at the reference oil fraction $w_{\text{o,ref}} = 0.0195$. For an off-anchor oil fraction the existing reference-anchored composite closure is

$$W_{\text{h}} = \alpha_o W_{\text{h,native}} + \beta_o q_o(a_{\text{h}}), \quad \alpha_o = \frac{1-w_o}{1-w_{\text{o,ref}}}, \quad \beta_o = \frac{w_o - w_{\text{o,ref}}}{1-w_{\text{o,ref}}}, \quad q_o = 0.9635 a_{\text{h}}^{2.7036}. \tag{29}$$

The same weights give dry-composite heat capacity $\alpha_o c_{p,\text{native}} + \beta_o c_{p,\text{oil}}$. Thus β_o vanishes at the reference anchor and is a signed correction below it, as in the declared DT oil fraction. It is not an extra oil arm added to the reference total-meal isotherm (Sec. S1.2). Retained water follows the two-parameter modified Luikov form fitted to desorption data for desolventizer-outlet meal pooled with the low-temperature soybean-meal data of Pixton and Warburton (1975) (Luikov, 1975; Luz et al., 2006),

$$W_{\text{w}}(a_{\text{w}}) = \frac{A_1}{1 + A_2\, \ln(1/a_{\text{w}})}, \qquad \ln a_{\text{w}} = \frac{1 - A_1/W_{\text{w}}}{A_2}, \tag{30}$$

with a_{w} the water activity and A_1 (kg/kg) and A_2 (dimensionless) its two fitted parameters. The isotherm is temperature-independent in the nominal closure; the temperature-dependent form is carried as a mandatory replacement-only caloric sensitivity case (Sec. S1.2). The shared prescribed-bath case continues this same analytic expression below the evidence anchor $W_{\text{w}} = 0.023$, for strictly positive water, retaining $A_1 = 0.88$, $A_2 = 12.184$ and the original upper capacity 0.236 kg/kg. This is a declared constitutive extrapolation, not new retention data. The analytic log activity is used directly when exponentiating it would underflow; water is neither floored nor held constant. Exactly absent water requires a separate component limit. Neither isotherm carries a cross-coupling term. The only direct co-sorption measurements support this at low water loading, but not beyond it (Cardarelli, 1998). Both isotherms are used outside their measured domains; these uses are marked as extrapolations in Section 5 rather than smoothed (closure detail in Sec. S6). Pure n-hexane follows the technical Helmholtz equation of state of Span and Wagner (2003a, 2003b), water the IAPWS-95 reference formulation (IAPWS, 2018; Wagner & Pruß, 2002), and the binary gas a second-virial mixture whose cross term carries a qualification interval rather than a value (Sec. S1.2). Table 2 gives the adopted material and transport inputs with their sources and scope. Published sorption coefficients retain their reported precision; derived quantities and declared transport inputs are rounded for presentation, while the calculations retain their full precision. Literature fits provide the thermodynamic properties; the remaining transport inputs define one common case. No coefficient was fitted to the falling-rate or DT outlet observations. The pore reference and its source-convention limits are stated with Eq. (12). For retained water, Touffet, Déléris, and Bokhove (2026) reports apparent moisture diffusivities reaching about 6.2×10^{-10} m^2/s in pelleted feed at 60 °C. We use that value as a reference response and derive the retained-water mobility from the coupled component balances at one reference state (Sec. S12). Its original value is held fixed in the pore-diffusivity study; it is not retargeted when D_b changes. This avoids assigning the same apparent

diffusivity independently to both water pathways. The material and temperature differences remain assumptions; neither coefficient is a measurement of soybean transport under DT conditions.

With the weights of Eq. (29), dry-matrix capacity is $c_{p,\mathrm{dm}}(T) = \alpha_o c_{p,\mathrm{native}} + \beta_o c_{p,\mathrm{o}}(T)$. The published rounded linear correlation for soybean oil of Garcia Rojas et al. (2013) (Table 7) gives

$$c_{p,\mathrm{o}}(T) = 1024 + 3T \quad \mathrm{J/(kg\,K)}, \tag{31}$$

$$u_{\mathrm{dm}}(T) - u_{\mathrm{dm}}(T_s) = \alpha_o c_{p,\mathrm{native}}(T - T_s) + \beta_o\big[1024(T - T_s) + 1.5(T^2 - T_s^2)\big]. \tag{32}$$

Temperatures are in kelvin and T_s is the common solid reference. Energy uses this integral, and its temperature derivative uses the same capacity. The reported experimental range is 299.15–433.15 K. At the native oil fraction, $\beta_o = 0$ and the oil correction cancels exactly. At the lower DT oil fraction it remains a small signed correction, without adding a second oil inventory.

The loading-dependent retained-water diffusivity is compared with oilseed-meal vapour-sorption and soaking measurements in Sec. S12. The liquid-connected coefficients are defined in Section 2.6; their earlier constant-property sensitivity is reported in Sec. S12. The separate older effective-diffusivity comparator is retained only in the supplement. The bed film follows the stated correlation convention of Faner (2008) as transcribed by Coletto et al. (2022) (Sec. S1.2). The ranges in Table 2 describe source evidence or separate sensitivities; the completed marches use the adopted values in its middle column. Section 4 gives the experiment-specific bath histories and initial loadings.

Table 2: Adopted property and transport inputs for the two particle experiments.

Input	Adopted value	Basis and interpretation
Particle radius R_{p}, mm	0.885 (Faner); 1.50 (DT)	Source mean sphere and coarse end of source size range, respectively (Faner et al., 2019)
Porosity ε_{g}	0.141	Soybean-meal measurement (Faner et al., 2019); fixed in both baths
Dry meal density $\rho_{\mathrm{dm,p}}$, kg/m^3	1.16×10^3	$(1-\varepsilon_{\mathrm{g}})\rho_s$; skeletal density $\rho_s = 1.35 \times 10^3$ kg/m^3 (Faner et al., 2019)
Reference critical loading, kg/kg dry	0.20	Source capacity anchor (Faner et al., 2019); event loading uses live liquid density
Critical-volume factor κ_v	2.675	Derived from the 0.20 anchor and reference liquid density 615 kg/m^3
Reference oil fraction $w_{\mathrm{o,ref}}$	0.0195	Anchor of the total-meal retention law; DT oil fraction is 0.0137
GAB X_m, kg/kg dry	5.183×10^{-3}	Published soybean sorption fit (Cardarelli & Crapiste, 1996)
GAB C_0, K_0	3.117×10^{-3}; 9.172×10^{-2}	Same fit; neither coefficient is adjusted here
GAB $\Delta H_C/R$, $\Delta H_K/R$, K	2262; 729.6	Published temperature dependence (Cardarelli & Crapiste, 1996)
Water isotherm A_1, kg/kg dry; A_2	0.880; 12.184	Modified Luikov fit (Luz et al., 2006; Pixton & Warburton, 1975)
Water capacity W_{cap}, kg/kg dry	0.236	Upper source-domain anchor; positive-water continuation below 0.0230 is declared
Meal and oil heat capacities, J/(kg K)	2.32×10^3; $1024 + 3T$	Meal prior (Coletto et al., 2022); oil: Eq. (31), T in K
Wet/dry conductivity, W/(m K)	0.24	Porous-mixture prior (Coletto et al., 2022); 0.29 is a separate solid-property sensitivity
Binary pore D_b, m^2/s	$D_{b,\mathrm{ref}} = 4.0 \times 10^{-10}$	Assumed reference at 105°C, 1 atm; source scale and local scaling: Eq. (12) and text
Retained-water mobility $L^{\mathrm{eff}}_{\mathrm{w,ret}}$, mol/(m s)	8.47×10^{-7}	Derived from pelleted-feed response (Touffet et al., 2026); conversion: Sec. S12
Added liquid-water mobility $\mathcal{L}_\ell$, mol/(m s)	8.39×10^{-5}	Hemmingsen size/time estimate: added $99L^{\mathrm{eff}}_{\mathrm{w,ret}}$, total $100L^{\mathrm{eff}}_{\mathrm{w,ret}}$; assumptions: Sec. S12
Liquid contact K_ℓ, mol/(m^2 s)	0.32	Conditional scale from hot-water uptake of soybean meal (Hemmingsen et al., 2008); construction: Sec. S12
Cross-virial interaction k_{wh}	0.494	Central correlation value; interval 0.477–0.511 (Plyasunov & Shock, 2003)
Pressure, kPa	101.3	Atmospheric reference; 101–170 is the wider process envelope (Kemper, 2013)
Faner film exposure multiplier	0.146	Constant-rate data (Faner et al., 2019), Fig. 2; calculation: Sec. S7.2

3 Numerical implementation and verification

This section explains how the particle equations are integrated in time and how accepted states are checked for admissibility and conservation. The numerical calculation advances component inventories and energy under the prescribed gas state, heat supply and exposure. Surface composition, particle temperature and front motion are solved responses. Each phase regime uses the corresponding equations of Section 2; the numerical scheme transfers the conserved quantities across regime changes.

3.1 Spatial discretization and coupled time integration

Regional balances are integrated in spherical finite volumes. In regime A, the present experiments represent the particle as a uniform thermal body, coupled to the liquid surface through a one-cell conductive resistance. This reduces the coupled startup problem to one body temperature, while retaining a separate surface temperature and conductive heat transfer. It is a coarse approximation, not a spatially resolved preheating calculation; its effect on the initial temperature profile should be assessed before detailed startup heating is interpreted. Regimes B and C resolve the radial particle. The receding interface cuts a fixed radial grid and is represented by the wet volume fraction $z = (s/R_\mathrm{p})^3$. Its swept-volume rate is

$$Q_\Gamma = V_{R_\mathrm{p}} \frac{z_\mathrm{new} - z_\mathrm{old}}{\Delta t}. \tag{33}$$

The same rate enters both regional balances and the interfacial jumps. Backward-Euler integration solves temperatures, compositions, external liquid inventories and front position together. Component enthalpy transport and conductive heat use the same face states as the storage calculation. Detailed face reconstruction is given in Sec. S2; history-specific controls are collected in Sec. S2.2.

3.2 Phase changes and moving-interface events

When external free hexane is depleted, the dry shell can appear from zero thickness. The first finite-volume interval satisfies the component and energy jumps without assigning an arbitrary initial shell. Arrival at and departure from radial grid faces are solved as geometric events. Near the centre, small positive-core intervals precede a conservative projection to the zero-core particle. That projection includes the external exchanges over its estimated duration; the conditional remaining-time allowance is below 2 ms for DT and 20 ms for Faner. These bounds concern the terminal event, rather than the error of the complete trajectory. External water appearance and depletion likewise switch the surface equations while preserving water and energy. The liquid-water imbibition pathway operates only in a dry shell connected to external liquid. The DT trajectory retains positive external water throughout regime B, so this experiment does not exercise the moving front without a water film. Regime C includes liquid-present, contact and liquid-absent surface states.

3.3 Physical admissibility and conservation checks

At each time step, the nonlinear solution must satisfy non-negative inventories, pressure-feasible phase equilibrium, the receding-front geometry and the retained-water capacity. Capacity is expressed as an active set,

$$0 \le W_\mathrm{cap} - W \;\perp\; \lambda \ge 0, \tag{34}$$

with λ its dimensionless dual variable. Local component, energy and combined inventory balances are checked at a normalized tolerance of 10^{-10}. Independent accumulation of the external

exchanges provides a separate whole-history check. The four-cell Faner reconstruction uses a reporting allowance of 4×10^{-10}, corresponding to at most 10^{-11} kg water/kg dry meal; its largest whole-history value is 3.9×10^{-10}, in water, and its step acceptance tolerance remains 10^{-10}. The DT history stays below 5×10^{-11} from dry-shell formation to exit and below 3.3×10^{-13} before it (Section 4.2).

Root-residual criteria, seed tolerances and conditional extinction-time bounds are distinct from conservation acceptance; their case-specific values are stated once in Sec. S2.2. The mesh and timestep comparisons reported with the results quantify selected sensitivities, not whole-history convergence (Section 5).

3.4 Verification of balances and numerical operators

The operator benchmarks retain their declared constant-property reference configurations. The temperature-dependent property extension is checked separately by the energy derivative and inverse-state identities, native wet and dry capacities, binary face-flux scaling and retained-water invariance; the two updated histories retain the native conservation gates.

This subsection checks that the discrete operators converge at their expected orders and that the discrete balances conserve water, hexane and energy. Manufactured solutions give spatial orders 1.95–2.00 and temporal orders 0.97–1.00 for the fixed-domain components; the moving-front time operator gives 0.98–0.99. These are tests of individual operators, not measured orders for an entire drying trajectory. Every step of the short refinement ladder satisfies its component and energy balances at or below 10^{-10} in physically normalized units (at most 7.8×10^{-13} on that ladder). The same swept volume in all balances enforces the discrete geometric conservation identity exactly. Shifting the arbitrary enthalpy zero of hexane does not change the solution: the observed datum-invariance difference is 3.42×10^{-10} under its separately stated environment allowance. The discrete identity (19) is the corresponding test for water. It is the check we recommend for any moving-boundary model that claims to count latent and binding heat once (Section 2.4). The full operator-by-operator table, an illustrative numerical sensitivity case that is not a valid limiting case at its initial state, and the unresolved tests are in Secs. S2 and S2.

The discrete caloric-datum test (19) also exposed a consequential interfacial error in an earlier implementation, now corrected. There, the material swept by the front was removed at the equilibrium interface composition but debited from the liquid-hexane core at its bulk composition. Both component and energy must use the same consumed material, as required by (15)–(16); the corrected comparisons use the wet-side material state. These numerical checks establish conservation of the stated equations, not the identity of their unmeasured transport coefficients or predictive accuracy in a plant. The front's trajectory refinement, including a nonconverged flux-integral result, is reported in Sec. S2.

4 Results

This section applies the particle model to two prescribed baths. The Faner experiment compares solvent loading and temperature with source readings; the DT experiment follows the coupled solvent, moisture and thermal response of a particle from feed to exit. Both use the same particle conservation laws, phase equilibrium, retention closures and reference transport coefficients; their gas histories, geometry and initial states differ. Liquid-water imbibition operates when an external water film contacts the dry shell.

4.1 Falling-rate drying in Faner's superheated-hexane experiment

Experimental basis and declared particle. The soybean run of Faner et al. (2019) supplies pure n-hexane vapour at 120 °C and atmospheric pressure. We represent its reported mean diameter by a sphere of radius 0.885 mm, initially at the calculated hexane boiling temperature, 68.8 °C. Initial loadings are $X_{\mathrm{h}} = 0.475$ and $X_{\mathrm{w}} = 0.0250$ kg/kg dry meal, with oil mass fraction 0.0195. The source reports preheating but neither its exact endpoint nor initial moisture; these are declared initial conditions. The sphere preserves equivalent volume, not flake shape or a mass-weighted size distribution. Source preparation, size descriptors, alternative mappings and measurement definitions are collected in Sec. S7.1.

Heat supply and shared model. The exposure multiplier 0.146 carries a boundary calibration from the source's early constant-rate duty, using an earlier size ensemble (Sec. S7.2). It scales heat and mass coefficients at fixed spherical area, giving $h_Q = 15.1$ rather than 104 W/(m^2 K). It was not reidentified for the present sphere. The heat duty does not independently identify mass transfer, so applying the same multiplier to both coefficients is a boundary assumption. Falling-rate and temperature readings are reserved for comparison, conditional on that boundary.

The separate 2008 constant-rate duty checks, including their alternative sample-mass conventions and departures from Whitaker's correlation, are reported in Sec. S3. An otherwise unchanged calculation with multiplier one exhausts external free hexane at 6.8 s, compared with 46.8 s in the reported calculation. This early-stage sensitivity demonstrates the importance of the gas-side boundary; it does not establish that the industrial reference coefficients represent Faner's apparatus (Sec. S7.2). For reference, Faner's geometric comparator is

$$s = R_{\mathrm{p}} \left(\frac{X_{\mathrm{h}} - X_{\mathrm{e}}}{X_{\mathrm{c}} - X_{\mathrm{e}}} \right)^{1/3}, \tag{35}$$

Here X_{e} is Faner's equilibrium hexane loading for the source's gas conditions. In contrast, $X_{\mathrm{e}}^{*}(T, P)$ in Section 2.1 is our retained-hexane reference evaluated at the pressure-feasible saturated activity a^{*}. Both have units kg hexane/kg dry meal, but they refer to different equilibrium specifications and are not interchangeable. Neither fixes the core-extinction loading in the present calculation, which solves the component and energy jumps; Eq. (35) is only the source's geometric comparator.

Both experiments use the constitutive laws and inputs of Section 2.8 and Table 2, without refitting particle coefficients. The regime-A water-access assumption is stated in Section 2.5; water evolves through both transport paths after shell formation. Imbibition is inactive in this experiment because no external water film forms.

Figure 2 compares the four-cell calculation with Faner's readings on the reported experimental clock. Loading includes all particle hexane per unit dry meal; temperature is the particle volume mean, while the thermocouple measures a local holder temperature. Markers show the measurements with their digitization half-widths. The dotted vapour line is the prescribed nominal temperature; the dashed line is the measured vapour trace, shown for comparison. Calculated water and liquid-hexane core volume are supporting results in Fig. S2.

Solvent removal and temperature. External free hexane disappears at 46.8 s; the core vanishes at 87.4 s. Loading crosses 0.20 kg/kg at 46.6 s and reaches 0.10 kg/kg a further 17.7 s later, versus 45.9 s and 25.4 s in the digitized source. The calculated falling interval is about 30% shorter. At 240 s, hexane is 0.00161 kg/kg dry, water is 0.0144 kg/kg dry and mean temperature is 114.4 °C. This discrepancy cannot be assigned to one mechanism: the particle sizes are known only as summaries, the initial moisture is unknown, and the local exposure and the thermocouple sampling are uncertain.

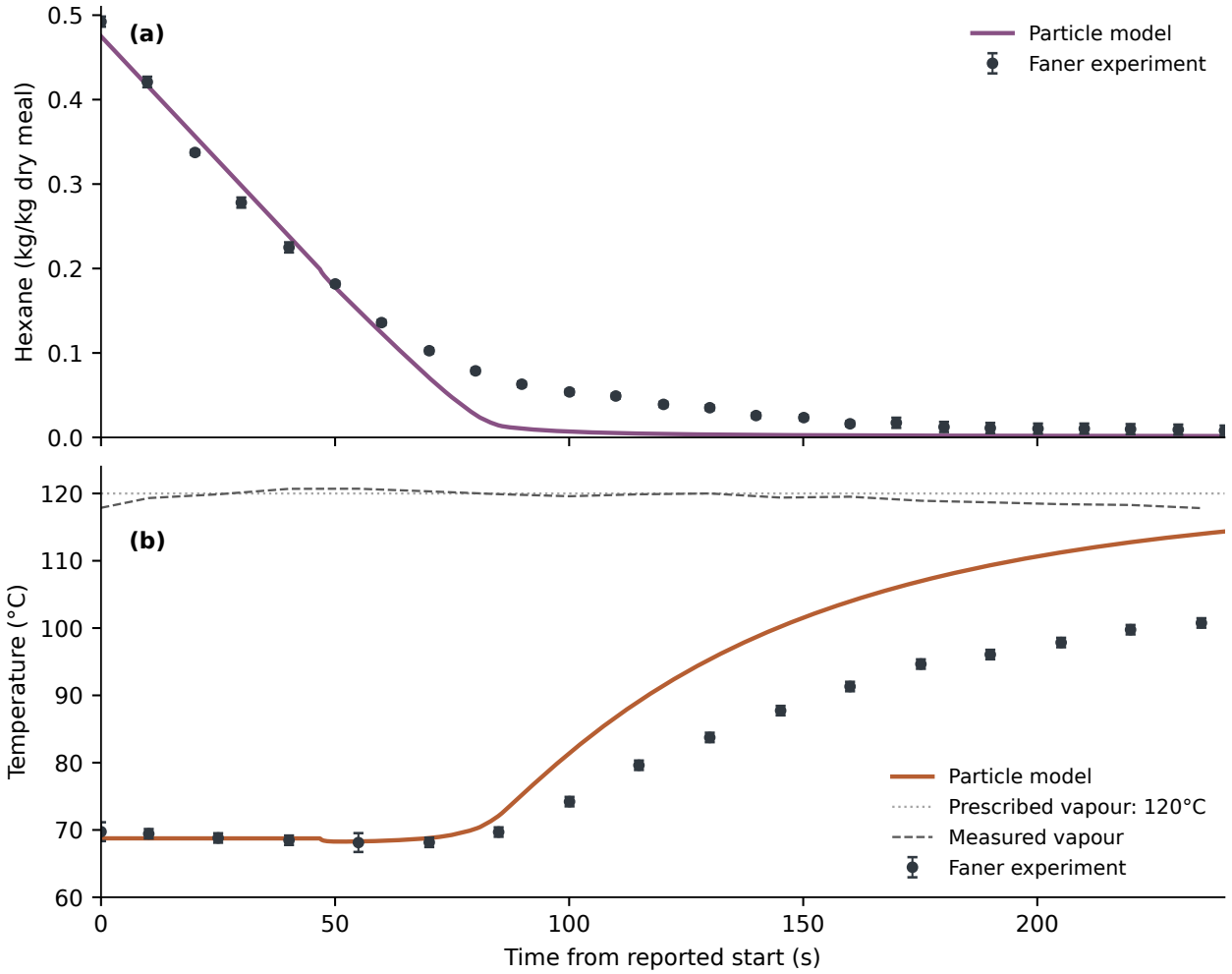


Figure 2: Faner's pure-hexane experiment: (a) hexane loading and (b) particle temperature. Lines show calculations; markers show measurements.

Table 3 summarizes the comparison; loading crossings are interpolated between successive calculated states. Its temperature comparison retains the distinction between local holder and particle mean.

Regime-C hexane transport in the pure-hexane bath. Decomposing the calculated hexane flux at the outer dry-shell face into total vapour motion and binary diffusion, as in Eq. (8), shows that hexane is carried outward by the moving pore gas. Binary diffusion contributes in the opposite direction throughout the recorded regime-C tail, offsetting 0.01–0.70% of the net outward hexane flux. These contributions describe how hexane moves in this calculation; they do not by themselves identify the rate-limiting resistance. Their reconstruction is given in Sec. S10.1.

Table 3: Faner comparison on the reported experimental clock.

Observable	Faner experiment	Particle model
Time to $X_h = 0.20$, s	45.9	46.6
Interval $0.20 \rightarrow 0.10$, s	25.4	17.7
Liquid-core disappearance, s	Not measured	87.4
Water at 240 s, kg/kg dry	Not reported	0.0144
Temperature at 235.1 s, °C	100.8	114.0

Earlier calculations with constant $D_b = 6 \times 10^{-10}$ m^2/s and oil capacity 2000 J/(kg K), using the reported mean diameter instead of the smaller sphere inherited from earlier calculations delays removal and warming: the external area per dry mass decreases and the internal transport distances increase. This improves the loading crossing while leaving a falling-rate discrepancy. At matched four-cell resolution, the 80–100°C temperature milestones occur about 12–17 s later than with the inherited smaller sphere. In a DT heat balance, that timing determines how long the heat supply goes mainly to evaporation before the meal warms, affecting residence requirements and product quality. The earlier source-mean two-cell march loses its core at 90.1 s; the earlier four-cell value is 88.1 s. Their positive-core step sequences also differ, so this comparison is a resolution sensitivity rather than a continuum limit.

The measured vapour varies from 117.8 to 120.7 °C, rather than remaining exactly at its nominal value. An earlier constant-property two-cell sensitivity applies that trace after core depletion and gives 113.0 °C at 240 s, compared with 114.3 °C under constant vapour temperature. The small improvement supports treating heat-supply variation as an experimental uncertainty; it does not close the full temperature discrepancy. Before core depletion, this sensitivity still uses the nominal bath (Sec. S5.2). Startup calculations at higher initial moisture, 0.050 and 0.100 kg/kg, are not yet resolved (Sec. S5).

4.2 Particle destiny in the prescribed DT bath

Bath and initial state. The DT experiment uses an effective spherical radius $R_\mathrm{p} = 1.5$ mm and follows the particle from feed to the prescribed exit at 1766.5 s (29.4 min). This radius is the coarse end of the previous size screening. It is consistent with the upper end of the 0.5–1.5-mm equivalent-radius range containing 95.1% of Faner's soybean particles (Faner et al., 2019), but is not a measured DT mean. The longer heat path and smaller surface area per unit mass make this a useful coarse-particle case when examining residence needed for solvent removal. It does not represent a mass-weighted meal size distribution.

Initial temperature is 49.0 °C, hexane and water loadings are 0.388 and 0.124 kg/kg dry meal, and oil mass fraction is 0.0137. The feed is cooler than the 57–60°C examples reported by Witte (1995) and Kemper (2013), so sensible heating remains part of this experiment. The selected temperature is a scenario input, rather than a measurement for the simulated charge. The pre-desolventizing (PD) bath is at 72.0 °C with hexane mole fraction 0.664, exposure fraction 0.00408 and indirect heat supply 85 W/kg dry until 490.2 s. Steam contact then leads to a toasting bath ending at 105.0 °C. The prescribed gas history is detailed in Sec. S11.

The sequence represents indirect pre-desolventizing followed by live-steam contact, as described by Kemper (2013). Indirect heating removes solvent before steam contact and reduces the solvent load on the steam-heated trays. The selected exposure fraction approximates the small exposed area of a bed compared with the summed areas of isolated spheres. This exposure fraction, the imposed duty and the bath ladder together define the experiment; they do not identify a measured industrial operating point. Liquid-water imbibition uses the contact and mobility estimates of Section 2.6.

Liquid depletion, condensation and imbibition. At PD exit, about 0.169 kg external free hexane/kg dry remains. It disappears approximately 2.1 s later while the external water film accumulates. The initial two-liquid surface equilibrium is near 61.3 °C; it differs from pure hexane's boiling point. The dry shell forms at 492.3 s and core extinction occurs at 503.5 s, approximately 11.3 s after external free-hexane depletion. Retained hexane and pore gas then continue to evolve in regime C. Both the rapid surface evaporation and the core recession follow from the coupled balances.

At core extinction, the particle contains 0.126 kg water/kg dry and the external water film contains a further 0.097 kg/kg dry. The film remains until 703.3 s, about three and a half minutes after dry-shell formation. Liquid-water imbibition and evaporation back to the bath compete for this inventory; condensation does not imply that all incoming water remains in the particle. The external water film supplies mass and energy for several minutes and persists into regime C.

Figure 3 presents wet-basis hexane and moisture alongside the volume-mean particle temperature. Dashed curves use the right axes for the prescribed bath mole fractions and temperature. The lower timeline identifies pre-desolventizing and main trays; dotted vertical lines mark A–B and B–C. The insets resolve the brief core recession and the external water film. Green markers at 30 min show published outlet ranges: 16–22% moisture from Witte and 100–500 ppm solvent and 105–110°C temperature from Kemper (Kemper, 2013; Witte, 1995). They provide industrial context rather than paired measurements of this prescribed bath. The calculated history ends at 29.4 min.

Regime-C hexane transport in the steam-rich bath. The two gas-phase contributions change within regime C. At the first recorded advance, 508.5 s, outward total vapour motion supplies 97.33% of the net outward hexane flux at the outer dry-shell face, and binary diffusion supplies 2.67%. From about 778 s to exit, total vapour motion is inward while hexane continues to leave the particle: outward binary diffusion transports hexane against the incoming gas. Inward gas motion partly offsets this outward diffusive flux. Diffusion is therefore essential to the computed late DT removal pathway (Sec. S10.1).

Exit observables and numerical sensitivity. Particle water at exit is 0.212 kg/kg dry meal, equivalent to 17.5% wet-basis moisture. Hexane is 168 mg/kg dry, or 138 ppm on a wet basis, and volume-mean particle temperature is 105.0 °C. The external water film is depleted. The particle-only basis conversion uses $1 + X_{\mathrm{w}} + X_{\mathrm{h}}$; water in the external water film is reported separately while the film is present. Table 4 compares the result with the published outlet ranges. Calculated ppm uses wet meal; the source solvent range does not specify its analytical mass basis. The calculation uses two radial cells and a 5-s maximum regime-C timestep.

Table 4: DT exit results and published outlet ranges.

Observable	Particle model	Published range
Hexane, mg/kg dry	168	Mass basis not specified
Hexane, ppm wet	138	100–500
Particle moisture, % wet	17.5	16–22
Volume-mean temperature, °C	105.0	105–110
External water film, kg/kg dry	0	Not measured

The calculated moisture and hexane lie within the published ranges, and the temperature, 104.94°C before rounding, is close to the lower edge of its range. These ranges are industrial context, not a validation: the agreement cannot identify the imbibition coefficients or validate the prescribed bath. With the same temperature-dependent property laws, varying the prescribed reference pore diffusivity from 2 to 6×10^{-10} m^2/s changes exit moisture from 17.319 to 17.615% wet basis and residual hexane from 145.48 to 130.27 ppm wet basis. The moisture span is 0.296 percentage points and the hexane reduction is 10.45%. This interval is a prescribed sensitivity range; it does not measure uncertainty in the material coefficient (Sec. S13). With the nominal laws, factor 100 and unchanged $K_\ell = 0.32$ mol/(m^2 s), halving the regime-C maximum timestep from 5 to 2.5 s changes moisture by +0.012 percentage points, hexane by −0.29%, temperature by

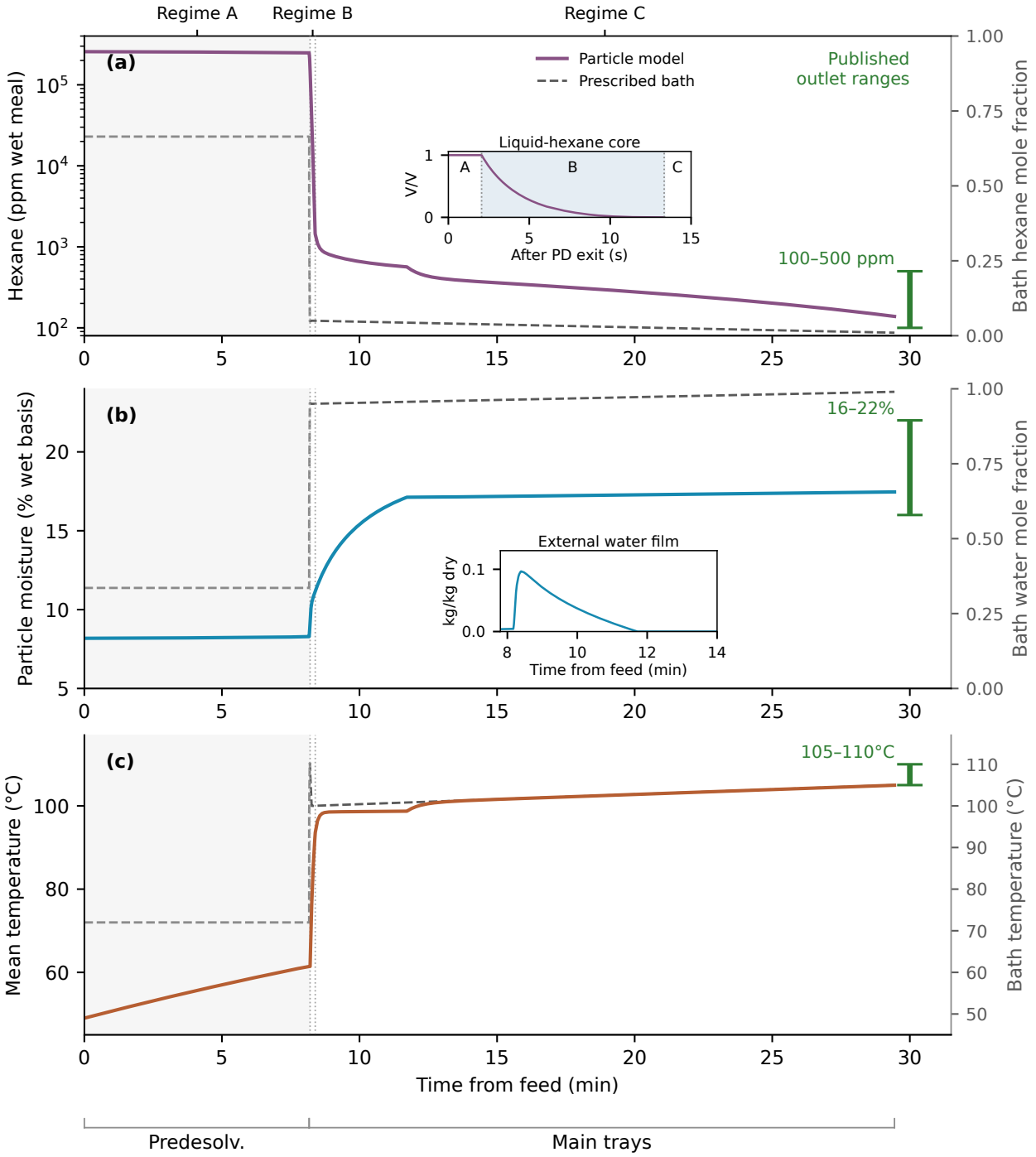


Figure 3: DT particle destiny: (a) hexane, (b) moisture and (c) temperature. Solid lines show the particle response; dashed lines show the bath. Green markers indicate published outlet ranges.

-0.00063 K and first recorded water-film depletion by -1.08 s. Both continuations start from the same accepted regime-B checkpoint and pass the independent component and energy audit. The prescribed diffusivity-range effect is larger than this regime-C timestep change; full-history mesh and time-step convergence is not yet established (Section 5).

An independent reconstruction of every accepted advance from dry-shell formation to exit gives maximum normalized water, hexane and energy defects of 8.31×10^{-12}, 2.37×10^{-12} and 2.34×10^{-12}, respectively. An independent exterior-rate reconstruction of the new feed history up to dry-shell activation stays below 3.76×10^{-13}. History-specific numerical controls and recoveries are collected in Sec. S2.2. Conservation of this discrete result does not identify the material transport rates.

5 Discussion

The two prescribed-bath experiments show how a common particle formulation connects external-liquid evaporation, internal-core recession and residual-component transport. Their signed component and energy exchanges can supply a future tray calculation. Prescribing the bath isolates this particle response, while leaving countercurrent vapour, wall, residence and discharge balances to the equipment model. We first interpret the Faner comparison, then its meaning for DT practice and what is needed next.

The calculated Faner particle dries faster than the measured charge. The setup uncertainties collected in Sec. S7.1 prevent assigning that discrepancy to one mechanism. Larger particles can prolong removal even when small particles heat earlier, but their contribution requires mass weighting and completed class histories (Sec. S5.1). Likewise, a subcooled start requires a new initial-value calculation with sensible heating, rather than a time shift of the temperature curve. Simultaneous loading, spatial temperature and size-resolved observations would distinguish startup, heat exposure and internal transport effects.

For DT practice, the three regimes divide the heat demand. Indirect heating in PD removes external free hexane before live-steam contact; steam condensation then supplies heat for rapid solvent removal, and the remaining retained hexane determines the later approach to the outlet residual. Industrial DT residence and heat supply must serve these successive needs (Kemper, 2013). At the same time, the moist heat treatment must reduce antinutritional activity while preserving useful protein quality (Mustakas et al., 1981; Witte, 1995). The calculated temperature and moisture histories describe that exposure; a meal-quality kinetic model is outside the present scope.

The moisture result also matters for energy design: water retained at DT exit contributes to the load on the downstream dryer, which takes meal towards its storage moisture (Witte, 1995). More condensation can accelerate heat supply and solvent removal, while more retained water can increase later drying duty. A steam-saving claim therefore requires the full balance of steam, indirect heat and drying energy, rather than outlet solvent or moisture alone. These prescribed-bath experiments expose that coupling without quantifying a plant steam saving.

Two independent equilibrium checks give further context for DT practice. At literature-motivated last-tray compositions, the dry-shell hexane equilibrium floor is 4.5–66.0 ppm without the separate oil-solution sensitivity and 8.1–112.4 ppm with it (Sec. S6). These equilibrium quantities do not set the residence needed to reach them. For retained water, a separate ideal-vapour diagnostic defines T^* by

$$a_{\mathrm{w}}(W_{\mathrm{w}})P^{\mathrm{w}}_{\mathrm{sat}}(T^*) = P. \tag{36}$$

Here W_{w} is retained-water loading, excluding the external water film. The product on the left is its approximate equilibrium water vapour pressure. Equating it to total pressure is a water-

only diagnostic; the actual binary particle calculation instead matches water fugacity to the water component of the gas. Equation (36) neither pins the transient particle temperature nor establishes boiling in the DT. Applying the earlier supplied-moisture diagnostic under the assumption that this moisture is retained gives 100.5–107.6 °C. It combines observations from different sources; at the moisture of the temperature study, the diagnostic temperature exceeds the observation by 2.1 K (Sec. S6).

A separate free-water droplet has unit water activity and, neglecting capillary-pressure effects, boils near 100 °C at 101.3 kPa (IAPWS, 2018; Wagner & Pruß, 2002). Its boiling temperature cannot be raised by assigning the lower activity of retained water elsewhere in the particle. Our retained-water inventory does not resolve molecularly bound and interstitial water, so bulk moisture alone cannot establish that all water has the isotherm activity. The 100.5–107.6 °C interval is therefore a conditional equilibrium diagnostic, not evidence of a measured boiling-point elevation of free liquid in meal. Neither this diagnostic nor agreement with industrial outlet ranges validates the coupled transient particle or the industrial tower.

Hexane leaves retained storage, enters the pore gas through local equilibrium and is carried outward by binary diffusion and total vapour motion. Retained hexane has no separate radial liquid flux; water also moves through the matrix (Section 2.2). These paths continue after core extinction without a fitted particle-mean tail law. For DT, the additional imbibition pathway and its material transfer assumptions are specified in Section 2.6. Because the water film persists into regime C, its assumed transparency to hexane may materially affect solvent removal. Transient uptake measurements on extracted meal, with external liquid quantified separately and several particle sizes, are needed to distinguish contact and interior resistances. An outlet moisture match alone cannot identify them.

Material and numerical limits set the present scope. The retention extensions and fluid-property domains are stated with their closures in Section 2.8; the effective pore reference and fixed-hindrance assumption accompany Eq. (12). These transport inputs remain to be identified from paired water–hexane measurements. Fixed geometry, an immobile connected core and the isobaric total-flux closure restrict the represented mechanisms (Sections 2.1–2.3). The two-cell DT and four-cell Faner histories are complete discrete experiments; operator verification and selected timestep checks do not yet establish whole-history continuum convergence.

6 Conclusions

One conservative particle formulation completes both a 240-s water-bearing particle experiment in pure superheated hexane and a 1766.5-s prescribed DT history. Component and energy balances continue through external-liquid depletion, internal-front recession and core extinction. External condensate is stored separately and exchanges water and energy with the particle and the prescribed gas. Both histories use temperature-dependent soybean-oil heat capacity with its consistent energy integral and a Fuller-shaped binary pore diffusivity. The oil correlation is measurement-based; the effective pore reference and its assumed fixed hindrance remain to be validated for solvent-wet meal.

The declared Faner particle falls from 0.20 to 0.10 kg hexane/kg dry meal in 17.7 s, compared with 25.4 s measured. The DT calculation includes liquid-water imbibition and reaches 17.5% wet-basis moisture, 138 ppm wet-basis hexane and a volume-mean particle temperature of 105.0 °C. Its liquid contact conductance is a conditional scale estimate from same-material uptake and its internal mobility is estimated from the soybean-meal size and uptake timescale. These transferred estimates are not jointly identified from DT-meal measurements.

The gas-phase transport mechanism differs between the two computed regime-C histories. In the pure-hexane experiment, total vapour motion carries hexane outward and binary diffusion contributes a small opposing flux. In the late steam-rich DT history, binary diffusion carries hexane outward against inward total vapour motion. The phase regime alone therefore does not determine which gas-phase contribution carries solvent removal. These findings identify the flux pathways; a dominant flux contribution does not by itself identify the rate-limiting resistance.

The model's signed water, hexane and energy fluxes can be coupled to tray balances for DT design; in that coupled form the formulation is the particle-level basis for a DT digital twin that can shorten process design and support model-based control. The results connect the model's mechanisms to observable trajectories and show which independent geometry, uptake and temperature measurements are needed before it is used for predictive equipment calculations.

Author contributions (CRediT)

Tomáš Svoboda: Conceptualization; methodology; software; formal analysis; validation; writing – original draft; writing – review and editing. Marcela Sluková: Resources; supervision; writing – review and editing. Tomáš Moucha: Investigation; writing – review and editing. Phuong Hoang: Writing – review and editing. All authors have read and approved the submitted manuscript.

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Supplementary material

The submitted Supplementary Material, Secs. S1 to S15, gives the regional and interface derivation, property and benchmark sources, measured-curve scoring conventions, numerical acceptance and observed convergence results, including the unfavorable cases. An extended technical report retains the longer derivations, former full manuscript passages and research history; no main-text claim relies on that report alone.

Data availability

The governing balances, parameters, comparison conventions and numerical acceptance criteria are stated in the text, the tables and the submitted Supplementary Material. The particle-model source, the scripts that produced the two particle histories and the sensitivity cases, the run records behind the figures and the headline numbers of the main text, and the LaTeX sources are released at `https://github.com/tomas3svoboda/dt_particle_model` (submission snapshot `submission-2026-10-15`, with a SHA-256 manifest of every file). The release's number check binds the printed numbers to the released records and names those that rest on records the release does not hold; such records, among them the authors' working notes and the independent balance record of the DT feed history, are available from the corresponding author on request. A preservation archive with a persistent identifier is being prepared. Digitized readings of published figures are shared subject to the publishers' redistribution terms, with reading half-widths and their sources. The steady equipment-model calculation that supplies the prescribed bath of Section 4.2 is not part

of the release, but its output is. Numerical material that does not yet exist is listed in Section 5 and Sec. S6; it is not described as available.

Declaration of generative-AI assistance

The authors used AI assistants for language editing, code review and manuscript organization. All physical assumptions, equations, numerical results, claim boundaries and conclusions were formulated, checked and are owned by the authors.

Conflict of interest

Tomáš Svoboda is affiliated with the University of Chemistry and Technology Prague and with Siemens, s.r.o., of which he is an employee; Siemens, s.r.o. had no role in the design of the study, the analysis or the decision to publish. The other authors declare no competing interests.

Nomenclature

Symbols of the equations of this paper, with their SI units; a symbol whose unit depends on the context gives both. Loadings are per kilogram of dry, water- and solvent-free composite meal. Symbols used only in the Supplementary Material are listed there.

Latin symbols

A_1 capacity parameter of the modified Luikov water isotherm (30); kg/kg

A_2 shape parameter of the modified Luikov water isotherm (30); dimensionless

a^* pressure-feasible n-hexane activity (1); dimensionless

a_h n-hexane activity; dimensionless

a_w water activity; dimensionless

$\bar{h}_k$ partial molar enthalpy of a transported component on the common caloric datum; $\bar{h}_{\mathrm{h,d}}$ that of n-hexane on the dry side; J/mol

B Darcy permeability of the particle pore space (10); m^2

B_h binding potential of n-hexane, per kilogram of dry meal (6); J/kg

C GAB energy constant in (28); dimensionless

c molar density of the pore gas, in $L^{\mathrm{eff}}_{\mathrm{wh}} = c\, y_\mathrm{w} y_\mathrm{h} D_b$; $\mathrm{mol/m}^3$

$c_{\mathrm{h,g}}$ n-hexane mass concentration in the pore gas; $\mathrm{kg/m}^3$

c_p specific heat capacity at constant pressure; J/(kg K)

$C^{\Gamma}_{\mathrm{h,d}}$, $C^{\mathrm{b}}_{\mathrm{h,w}}$ n-hexane concentration at the dry-side equilibrium interface trace, and in the wet-side donor state (17); $\mathrm{mol/m}^3$

C_k total component amount in all local forms per unit particle volume; $\mathrm{mol/m}^3$

c_w constant shift of the caloric zero of water (19); J/mol

D_{eff} effective diffusivity of n-hexane in the separate Fickian comparison, Eq. (S2); m^2/s

D_b binary pore coefficient of the mobility $L^{\mathrm{eff}}_{\mathrm{wh}}$; m^2/s

d_p mean equivalent particle diameter in the dispersion estimate; m

e_w specific energy of the liquid-hexane core, per kilogram of dry meal (6); J/kg

f_{exp} fraction of particle surface exposed to the external gas film in the prescribed DT bath experiment; dimensionless

h, a, ha external film coefficient per unit particle area; specific external area per dry mass; and their product, the film supply per kilogram of dry meal; $\mathrm{W/(m}^2\,\mathrm{K)}$; m^2/kg; W/(kg K)

h_{dm} specific enthalpy of the dry composite meal; J/kg

ΔP pore-gas overpressure across the dry shell (10); Pa

$h_{\mathrm{w,l}}$ specific enthalpy of liquid water; J/kg

j_{dm}, j_{L_o} fluxes of the immobile dry meal and residual oil, zero in (3); $\mathrm{kg/(m}^2\,\mathrm{s)}$

J_E total radial energy flux; $\mathrm{W/m}^2$

J_i binary exchange flux of component i in the pore gas, $J_\mathrm{w} = -J_\mathrm{h}$ (7); $\mathrm{mol/(m}^2\,\mathrm{s)}$

j_k radial flux of component k; $\mathrm{mol/(m}^2\,\mathrm{s)}$

$\mathcal{L}_\ell$, K_ℓ additional retained-water shell mobility and finite liquid-contact conductance; mol/(m s); $\mathrm{mol/(m}^2\,\mathrm{s)}$

K GAB constant (28); dimensionless

k component index, $k \in \{\mathrm{w}, \mathrm{h}\}$

k_C regime-C particle-mean hexane relaxation rate, $15 D_{\mathrm{eff}}/R_\mathrm{p}^2$; 1/s

k_{dry} thermal conductivity of the dry shell; W/(m K)

k_m packed-bed external gas-film mass-transfer coefficient in the prescribed DT bath experiment; m/s

L half-thickness of a plane flake; m

$L^{\mathrm{eff}}_{\mathrm{wh}}$ binary pore mobility (7); L^{-}_{wh}, L^{+}_{wh} its lower and upper bound; mol/(m s)

$L^{\mathrm{eff}}_{\mathrm{w,ret}}$ retained-matrix water mobility (8); $L^{-}_{\mathrm{w,ret}}$, $L^{+}_{\mathrm{w,ret}}$ its lower and upper bound; mol/(m s)

M_w molar mass of water; kg/mol

N^{g}_i pore-gas molar flux of component i (8); $\mathrm{mol/(m}^2\,\mathrm{s)}$

N_t total (Stefan) molar flux of the pore gas; $\mathrm{mol/(m}^2\,\mathrm{s)}$

$N_{\mathrm{w,ret}}$ retained-matrix water flux (8); $\mathrm{mol/(m}^2\,\mathrm{s)}$

$N^{\mathrm{film}}_\mathrm{w}$ outward-positive water molar flux across the external gas film in the prescribed DT bath experiment, negative for condensation; distinct from $N_{\mathrm{w,ret}}$; $\mathrm{mol/(m}^2\,\mathrm{s)}$

P total pressure; Pa

$P^{\mathrm{h}}_{\mathrm{sat}}$, $P^{\mathrm{w}}_{\mathrm{sat}}$ vapour pressure of n-hexane and of water; Pa

q radial conductive heat flux; $\mathrm{W/m}^2$

Q_Γ swept-volume rate of the front (33); m^3/s

R, R_g molar gas constant; J/(mol K)

r radial coordinate; m

$r_1(t)$, $r_2(t)$ moving inner and outer radii of a spherical control volume in (13)–(14); m

r_{pore} pore radius in the capillary-bundle permeability; m

R_E, R_w discrete energy and water residuals of the interface rows (19); W; mol/s

R_p particle radius; m

s radius of the evaporation front; m

$\dot{s}$ front velocity, negative while the front recedes; m/s

T temperature; K

t time; s

Δt time step; s

$T^{\mathrm{bp}}_{\mathrm{hex}}$ boiling temperature of n-hexane in the flashing-zone interpolation of Coletto et al. (2022); K

T_L flashing-zone solid temperature in that interpolation; K

T_V vapour temperature in that interpolation; K

T_g gas temperature; K

T_Γ interface temperature; K

T^* temperature of the retained-water vapour-pressure diagnostic (36); K

u total local internal energy density; $\mathrm{J/m}^3$

u^{Γ}_{d}, $u^{\mathrm{b}}_{\mathrm{w}}$ u at the dry-side equilibrium interface trace and in the wet-side donor state; $\mathrm{J/m}^3$

$u_{\mathrm{h,l}}$ specific internal energy of liquid n-hexane; J/kg

V_{R_p} particle volume; m^3

W retained-water loading in (34); kg/kg

W_{cap} retained-water capacity, the upper end of the measured water evidence; kg/kg

W_{eq} equilibrium retained n-hexane loading in Eq. (S2); kg/kg

W_{native} retained n-hexane on reference soybean meal (28); kg/kg

w_{o} residual-oil mass fraction of the dry meal; dimensionless

$w_{\mathrm{o,ref}}$ residual-oil mass fraction of the reference meal, the anchor of the total-meal isotherm; dimensionless

w_{h} weight on the n-hexane boiling point in the flashing-zone interpolation of Coletto et al. (2022); dimensionless

W_{w} retained-water loading of the Luikov isotherm (30); kg/kg

x reduced activity Ka_{h} of the GAB isotherm (28); dimensionless

X_{c} critical hexane loading at external free-hexane exhaustion; kg/kg

X_{e} equilibrium hexane loading at specified gas conditions; Faner's value in Eq. (35); kg/kg

X_{e}^{*} retained-hexane equilibrium reference at pressure-feasible saturated activity; kg/kg

X_{f} external free-hexane mass per unit dry meal mass; kg/kg

X_{h} total n-hexane loading; kg/kg

$X_{\mathrm{h,w}}^{a}$ apparent mobile-liquid n-hexane loading of the liquid-hexane core, set at activation and carried with the solid; distinct from the total loading X_{h} and from retained hexane; kg/kg

X_m GAB monolayer loading (28); kg/kg

X_{w} total water loading; kg/kg

$Y_{\mathrm{w},b}$, $Y_{\mathrm{w},I}$ bulk and particle-surface gas mass fractions of water in the prescribed DT film closure; dimensionless

y_i mole fraction of component i in the pore gas; dimensionless

$y_{\mathrm{w},I}$, $y_{\mathrm{w},b}$ particle-surface and bulk-gas water mole fractions in the prescribed DT film closure; dimensionless

z wet-volume fraction $(s/R_{\mathrm{p}})^3$; z_{new}, z_{old} at the new and old time level; dimensionless

Greek symbols

α physical region label, w (liquid-hexane core) or d (dry shell); not a cell index

β_{A} Ackermann blowing argument of the film law; dimensionless

ε_{g} gas-filled porosity of the dry shell; dimensionless

θ_a temperature of a material shell at activation; K

λ multiplier of the capacity condition (34), a contribution to $\mu_{\mathrm{w,ret}}/(RT)$ carrying no mass, volume, enthalpy or internal energy; dimensionless

μ_g pore-gas viscosity (10); Pa s

μ_{w}, μ_{h}, $\mu_{\mathrm{w,ret}}$ chemical potential of water and of n-hexane in the pore gas, and of retained water; J/mol

ρ gas mass density in the dispersion estimate, $(\rho c_p)_{\mathrm{g}}$; kg/m^3

ρ_f gas mass density used by the external film pair in the prescribed DT bath experiment; kg/m^3

$\rho_{\mathrm{dm,p}}$ density of the dry composite meal in the particle; kg/m^3

τ pore tortuosity in the capillary-bundle permeability; dimensionless

ϕ_i fugacity coefficient of component i, omitted by the ideal-gas closure; dimensionless

Subscripts and superscripts

a at activation

b donor state, the wet-core material the front sweeps (superscript); on $y_{\mathrm{w,b}}$, the bulk gas

c critical

cap capacity

d dry shell

dm dry composite meal

e, eq equilibrium; eq on d: equivalent

eff effective

f external free hexane

g gas, pore gas

h n-hexane

i, k component index

l liquid

m monolayer

native reference soybean meal, at its own oil content

new, old new and old time level

o residual oil

p particle

ref reference

ret retained (sorbed) phase

s surface; on D_s, the solid

sat saturation

vap vaporization

w water; as a region label (on $[\![\cdot]\!]$, $C_{\mathrm{h,w}}$, u_{w}), the liquid-hexane core

Γ interface; as a superscript, the equilibrium interface trace

α region, w or d

* pressure-feasible equilibrium reference (on a and X_{e}); retained-water diagnostic (on T)

$^{-}$, $^{+}$ lower and upper bound of an interval

Operators and notation

$[\![\cdot]\!]_{\mathrm{w}}^{\mathrm{d}}$ jump across the front, dry-side value minus wet-side value

Δ a difference or a change, as in a time step or an enthalpy of vaporization

$\mathrm{D}_s/\mathrm{D}t$ time derivative following the solid; 1/s

∇ radial gradient; 1/m

∂ partial derivative

$\dot{x}$ time derivative of x

$\perp$ complementarity: both sides non-negative and their product zero

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